

1. Administrative & Study Identification

Full Study Title: A Global Study of Midostaurin in Combination With Chemotherapy to Evaluate Safety, Efficacy and Pharmacokinetics in Newly Diagnosed Pediatric Patients With FLT3 Mutated AML **Short Title/Acronym:** Midostaurin in Pediatric FLT3-Mutated AML **Protocol Number:** CPKC412A2218 (2017-004830-28) **NCT Number:** NCT03591510 **Posting ID:** 708149944546872987 **Sponsor Name:** Novartis (Novartis Pharmaceuticals) **Investigational Product:** Midostaurin (PKC412) **Phase of Development:** Phase II **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the recommended phase II dose (RP2D) of midostaurin in combination with chemotherapy based on the occurrence of dose-limiting toxicities (Part 1), and to evaluate the safety, tolerability, and efficacy of midostaurin combined with chemotherapy followed by 12 cycles of midostaurin post-consolidation therapy in pediatric patients (Part 2). **Secondary Objectives:** To evaluate the percentage of participants with response (CR, CR/modified CRi, CR/CRi), Time to Response (TTR), response duration, Event-Free Survival (EFS), Overall Survival (OS), and the pharmacokinetics (PK) of midostaurin.

2.2 Background & Rationale In pediatric patients with Acute Myeloid Leukemia (AML), the presence of an FLT3 mutation is indicative of high-risk disease, conferring a worse prognosis and a greater chance of relapse. While midostaurin—a multikinase inhibitor with potent activity against mutated FLT3—is approved and significantly improves survival in adults, this study addresses the unmet medical need for pediatric patients by testing its safety and efficacy in combination with standard chemotherapy for children with newly diagnosed FLT3-mutated AML.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 22 pediatric patients **Number of Sites:** Global, multi-center **Estimated Study Start:** 2018 **Estimated Primary Completion:** 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Ages 3 months to < 18 years of age. 2. Documented diagnosis of previously untreated de novo AML according to WHO 2016 criteria. 3. Presence of a FLT3 mutation status as measured/confirmed by a designated lab with results available prior to the first dose of midostaurin. 4. Lansky or Karnofsky performance status equal or superior to 60. 5. Adequate organ function: AST and ALT $\leq 3 \times$ ULN; Serum Total bilirubin $\leq 1.5 \times$ ULN; Estimated creatinine clearance ≥ 30 ml/min.

4.2 Exclusion Criteria 1. Any concurrent malignancy, AML with Philadelphia Chromosome, AML associated with Down syndrome (AML-DS), or JMML. 2. Symptomatic leukemic CNS involvement. 3. Isolated extramedullary leukemia, secondary AML and MDS. 4. Acute Promyelocytic Leukemia with the PML RARA rearrangement. 5. Prior treatment with a FLT3 inhibitor (up to 1 week of a FLT3 inhibitor, except midostaurin, prior to study enrollment is permissible).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Starting dose in Part 1 is 30 mg/m² twice daily (BID) (or 1 mg/kg bid for participants <10 kg body weight), escalated to 60 mg/m² BID based on Dose-Limiting Toxicities (DLTs) to determine the RP2D. The RP2D is used in Part 2. **Route of Administration:** Oral **Duration of Treatment:** Maximum planned duration is 17 cycles (5 blocks consisting of 2 induction and 3 consolidation blocks, followed by 12 cycles of continuous midostaurin post-consolidation therapy).
* **Block 1 (Both Parts):** Local standard induction (8-12 days) + Midostaurin for 14 days. * **Part 1 Blocks 2-5:** Block 2 FLADx (D1-D6); Block 3 HAM (D1-D4); Block 4 HA3E (D1-D5); Block 5 HiDAC (D1-D3). Midostaurin given D8-D21. * **Part 2 Blocks 2-5:** Block 2 HAM (D1-D4); Block 3 HA3E (D1-D5); Block 4 HAM (D1-D4); Block 5 HiDAC (D1-D3). Midostaurin given D8-D21.

5.2 Concomitant & Prohibited Therapy Permitted: Standard induction and consolidation chemotherapy including cytarabine (trade names: Alexan, Cytosar, DepoCyt, Tarabine PFS, Vyxeos; ATC: L01BC01), fludarabine, daunorubicin, mitoxantrone, etoposide. **Prohibited:** Other FLT3 inhibitors or concurrent investigational anti-leukemic agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Days -14 to 0 Informed Consent, WHO AML Diagnosis, FLT3 Mutation confirmation, Baseline Labs
Baseline	Day 1	Block 1 Induction	Chemotherapy initiation
Interim	Blocks 1-5	Chemotherapy duration dependent on block	

(D1-D6 max), Sequential Midostaurin D8-D21 per block;
Hematopoietic recovery & remission assessment by max Day 42 of
each block | | **End of Study** | Post-Consolidation | 12 cycles (28
days/cycle) of continuous midostaurin; long-term follow up for
Event-Free Survival (18 months min) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Part 1 - Occurrence of dose limiting toxicities (DLT):** Defined as any death due to toxicity related to study treatment (chemotherapy + midostaurin) and/or any CTCAE grade 4 non-hematological adverse event or abnormal laboratory value grade 4 related to study treatment unless the event improves sufficiently by day 42 and therefore does not further delay the next cycle.
- Part 2 - Safety of Midostaurin:** Evaluate the safety profile including type, frequency, and severity of adverse events during the induction, consolidation, and post-consolidation phases, as well as the post-treatment follow-up phase.
- Part 2 - Tolerability of Midostaurin:** Assessed by the number of dose interruptions/reductions and discontinuations due to the study drug.

7.2 Secondary & Exploratory Endpoints

- Percentage of participants with response (CR, CR/modified CRi, and CR/CRi).
- Time to response (TTR) and response duration.
- Event-Free Survival (EFS).
- Pharmacokinetics (PK) of midostaurin.
- Overall Survival (OS) (Exploratory).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical and laboratory monitoring via patient diary and HCP assessment. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for severe hematological and non-hematological toxicities. **Data Monitoring Committee (DMC):** An internal committee utilizing a Bayesian hierarchical logistic regression model guided by escalation with overdose control (EWOC) principle for dose escalation and safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for DLT and AE assessments; Intent-to-Treat (ITT)/Efficacy Set for survival outcomes. **Sample Size Justification:** A small cohort size of 22 patients in cohorts of 3 utilizing Bayesian hierarchical logistic regression for dose-finding (Part 1) and subsequent safety/efficacy confirmation (Part 2). **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimation) with appropriate censoring rules for patients lost to follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine global sponsor monitoring and site visits by Novartis Pharmaceuticals. **Audit Trail:** Secure electronic Case Report Form (eCRF) utilizing verified data clarification and locking procedures.

11. Study Identifiers & Governance

Primary ID: CPKC412A2218 **Posting ID:** 708149944546872987 **Registry Number:** NCT03591510 **Sponsor/Lead Organization:** Novartis **Study Title:** A Global Study of Midostaurin in Combination With Chemotherapy to Evaluate Safety, Efficacy and Pharmacokinetics in Newly Diagnosed Pediatric Patients With FLT3 Mutated AML **Official Regulatory Title:** A Phase II, Open-label, Single Arm Study to Evaluate the Safety, Efficacy, and Pharmacokinetics of Twice Daily Midostaurin (PKC412) Combined With Standard Chemotherapy and as a Single Agent Post-consolidation Therapy in Children With Untreated FLT3-mutated AML

12. Clinical Framework (AML Specific)

Primary Condition: FLT3-mutated Acute Myeloid Leukemia (Preferred UMLS: Acute Myeloid Leukemia with FLT3/ITD Mutation) **Disease Codes:** MeSH: D015470 | SNOMED: 107621000119106 **Diagnosis Threshold:** Documented de novo AML with confirmed FLT3 mutation **Medical Methodology:** Standard Induction/Consolidation Chemotherapy + Tyrosine Kinase Inhibitor **Optimization Target:** FLT3 inhibition targeting improved Event-Free Survival (EFS) and OS

13. Study Procedures & Methodology

Management Pathway: 5 blocks of systemic chemotherapy + 12 cycles of maintenance midostaurin monotherapy **Education Protocol (HCP):** Investigator protocol training on dose escalation and DLT identification **Education Protocol (Patient):** Informed consent (assent where applicable) and patient dosing diary for oral midostaurin **Quality Audit Mechanism:** Centralized laboratory confirmation of FLT3 mutation and DLT review committee for dose optimization

14. Observation & Follow-Up Schedule

Total Duration: 17 treatment cycles + at least 18 months of post-treatment follow-up **Onsite Visit Cadence:** Day 1 of each

chemotherapy block and active monitoring through D21 **Remote Monitoring Frequency:** Routine safety follow-ups as defined by institutional standards between cycles **Checklist Review Interval:** End of each treatment block (maximum D42) to assess hematopoietic recovery and CR/CRi status

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				